Problem 4

(a) SQL:

```
SELECT
    Ssn,
    Fname,
    Lname,
    Salary
FROM EMPLOYEE
WHERE Sex = 'F'
    AND Salary >= 25000
ORDER BY
    Salary DESC,
    Lname ASC;
```

Result:

ssn	fname	lname	salary
987654321	Jennifer	Wallace	43000.00
453453453	Joyce	English	25000.00
999887777	Alicia	Zelaya	25000.00

(b) SQL:

```
SELECT
    Fname,
    Lname,
    Bdate,
    Address
FROM EMPLOYEE
WHERE Bdate < DATE '1965-01-01'
    AND Lname LIKE '%e';
```

Result:

fname	lname	bdate	address
Jennifer	Wallace	1941-06-20	291 Berry, Bellaire, TX

(c) SQL:

```
SELECT
    P.Pnumber,
    P.Pname,
    P.Plocation,
```

```
    D.Dname,
    D.Mgr_ssn
FROM PROJECT P
JOIN DEPARTMENT D
    ON P.Dnum = D.Dnumber
WHERE D.Mgr_start_date >= DATE '1980-01-01';
```

Result:

pnumber	pname	plocation	dname	mgr_ssn
1	ProductX	Bellaire	Research	333445555
2	ProductY	Sugarland	Research	333445555
3	ProductZ	Houston	Research	333445555
10	Computerization	Stafford	Administration	987654321
20	Reorganization	Houston	Headquarters	888665555
30	Newbenefits	Stafford	Administration	987654321

(d) SQL:

```
SELECT
    P.Pnumber,
    P.Pname,
COUNT(W.Essn) AS EmployeeCount,
SUM(W.Hours) AS TotalHours
FROM PROJECT P
JOIN WORKS_ON W
    ON P.Pnumber = W.Pno
GROUP BY
    P.Pnumber,
    P.Pname
HAVING COUNT(W.Essn) >= 2;
```

Result:

pnumber	pname	employeecount	totalhours
10	Computerization	3	55.0
2	ProductY	3	37.5
20	Reorganization	3	25.0
1	ProductX	2	52.5
30	Newbenefits	3	55.0
3	ProductZ	2	50.0

(e) SQL:

```
SELECT DISTINCT
    E.Ssn,
    E.Fname,
    E.Lname,
    D.Dname
FROM EMPLOYEE E
JOIN WORKS_ON W
    ON E.Ssn = W.Essn
JOIN PROJECT P
    ON W.Pno = P.Pnumber
JOIN DEPARTMENT D
    ON E.Dno = D.Dnumber
WHERE E.Dno <> P.Dnum;
```

Result:

ssn	fname	lname	dname
333445555	Franklin	Wong	Research
987654321	Jennifer	Wallace	Administration

(f) SQL:

```
SELECT
    S.Ssn,
    S.Fname,
    S.Lname,
    COUNT(E.Ssn) AS DirectReportCount,
    AVG(E.Salary) AS AverageSalary
FROM EMPLOYEE S
JOIN EMPLOYEE E
    ON S.Ssn = E.Super_ssn
GROUP BY
    S.Ssn,
    S.Fname,
    S.Lname
ORDER BY
    DirectReportCount DESC,
    S.Ssn ASC;
```

Result:

ssn	fname	lname	directreportcount	averagesalary
333445555	Franklin	Wong	3	31000.000000000000
888665555	James	Borg	2	41500.000000000000
987654321	Jennifer	Wallace	2	25000.000000000000

(g) SQL:

```
SELECT
    D.Dnumber,
    D.Dname,
    COUNT(E.Ssn) AS EmployeeCount,
    AVG(E.Salary) AS AverageSalary
FROM DEPARTMENT D
JOIN EMPLOYEE E
    ON D.Dnumber = E.Dno
GROUP BY
    D.Dnumber,
    D.Dname
HAVING COUNT(E.Ssn) >= 3
ORDER BY
    AverageSalary DESC;
```

Result:

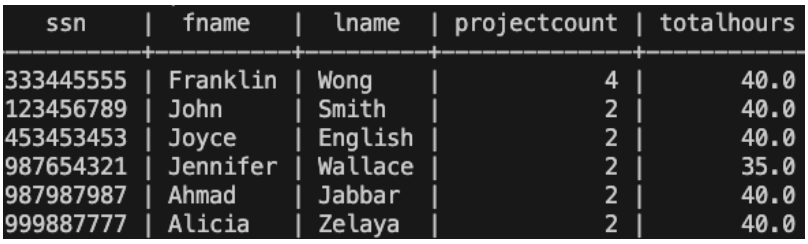
dnumber	dname	employeecount	averagesalary
5	Research	4	33250.000000000000
4	Administration	3	31000.000000000000

(h) SQL:

```
SELECT
    E.Ssn,
    E.Fname,
    E.Lname,
    COUNT(W.Pno) AS ProjectCount,
    SUM(W.Hours) AS TotalHours
FROM EMPLOYEE E
JOIN WORKS_ON W
    ON E.Ssn = W.Essn
GROUP BY
    E.Ssn,
```

```
    E.Fname,
    E.Lname
HAVING COUNT(W.Pno) >= 2
ORDER BY
    ProjectCount DESC,
    E.Ssn ASC;
```

Result:



ssn	fname	lname	projectcount	totalhours
333445555	Franklin	Wong	4	40.0
123456789	John	Smith	2	40.0
453453453	Joyce	English	2	40.0
987654321	Jennifer	Wallace	2	35.0
987987987	Ahmad	Jabbar	2	40.0
999887777	Alicia	Zelaya	2	40.0

Problem5

Table 1: Data Dictionary for `EMPLOYEE`

Column Name	Meaning	Data Type	Key	Constraint	Domain
`Fname`	Employee's first name	`VARCHAR(50)`		`NOT NULL`	String
`Minit`	Employee's middle initial	`CHAR(1)`			Single character
`Lname`	Employee's last name	`VARCHAR(50)`		`NOT NULL`	String
`Ssn`	Employee's social security number	`CHAR(9)`	`PK`	`NOT NULL, UNIQUE`	9-character SSN
`Bdate`	Employee's birth date	`DATE`			Valid date
`Address`	Employee's address	`VARCHAR(255)`			String
`Sex`	Employee's sex	`CHAR(1)`			M, F
`Salary`	Employee's salary	`NUMERIC(10, 2)`			Numeric value
`Super_ssn`	Supervisor's SSN	`CHAR(9)`	`FK: EMPLO`		Existing employee SSN

			YEE(Ssn)		
Dno	Department number	INTEGER	FK: DEPARTMENT(Dnumber)	NOT NULL, Default(1)	Existing department number

Referential Triggers

Foreign Key	On Delete	On Update
Super_ssn: EMPLOYEE(Ssn)	SET NULL	CASCADE
Dno: DEPARTMENT(Dnumber)	SET DEFAULT	CASCADE

Table 2: Data Dictionary for DEPARTMENT

Column Name	Meaning	Data Type	Key	Constraint	Domain
Dname	Department name	VARCHAR(50)		NOT NULL, UNIQUE	String
Dnumber	Department number	INTEGER	PK	NOT NULL, UNIQUE	Integer
Mgr_ssn	Department manager's SSN	CHAR(9)	FK: EMPLOYEE(Ssn)	NOT NULL, Default('888665555')	Existing employee SSN
Mgr_start_date	Manager's start date	DATE			Valid date

Referential Triggers

Foreign Key	On Delete	On Update
Mgr_ssn: EMPLOYEE(Ssn)	SET DEFAULT	CASCADE

Table 3: Data Dictionary for DEPT_LOCATIONS

Column Name	Meaning	Data Type	Key	Constraint	Domain
Dnumber	Department number	INTEGER	PK, FK: DEPARTMENT(Dnumber)	NOT NULL	Existing department number

Dlocation	Department location	VARCHAR(50)	PK	NOT NULL	String

Primary Key: (Dnumber, Dlocation)

Referential Triggers

Foreign Key	On Delete	On Update
Dnumber: DEPARTMENT(Dnumber)	CASCADE	CASCADE

Table 1: Data Dictionary for PROJECT

Column Name	Meaning	Data Type	Key	Constraint	Domain
Pname	Project name	VARCHAR(50)		NOT NULL, UNIQUE	String
Pnumber	Project number	INTEGER	PK	NOT NULL, UNIQUE	Integer
Plocation	Project location	VARCHAR(50)			String
Dnum	Controlling department number	INTEGER	FK: DEPARTMENT(Dnumber)	NOT NULL	Existing department number

Referential Triggers

Foreign Key	On Delete	On Update
Dnum: DEPARTMENT(Dnumber)	CASCADE	CASCADE

Table 5: Data Dictionary for WORKS_ON

Column Name	Meaning	Data Type	Key	Constraint	Domain
Essn	Employee's SSN	CHAR(9)	PK, FK: EMPLOYEE(Ssn)	NOT NULL	Existing employee SSN
Pno	Project number	INTEGER	PK, FK: PROJECT(Pnumber)	NOT NULL	Existing project number

Hours	Hours worked on the project	NUMERIC(5, 1)		Numeric value

Primary Key: (Pno, Essn)

Referential Triggers

Foreign Key	On Delete	On Update
Essn: EMPLOYEE(Ssn)	CASCADE	CASCADE
Pno: PROJECT(Pnumber)	CASCADE	CASCADE

Table 6: Data Dictionary for DEPENDENT

Column Name	Meaning	Data Type	Key	Constraint	Domain
Essn	Employee's SSN	CHAR(9)	PK, FK: EMPLOYEE (Ssn)	NOT NULL	Existing employee SSN
Dependent_name	Dependent's name	VARCHAR(100)	PK	NOT NULL	String
Sex	Dependent's sex	CHAR(1)			M, F
Bdate	Dependent's birth date	DATE			Valid date
Relationship	Relationship to employee	VARCHAR(100)			String

Primary Key: (Essn, Dependent_name)

Referential Triggers

Foreign Key	On Delete	On Update
Essn: EMPLOYEE(Ssn)	CASCADE	CASCADE